

Dense-intelligence research

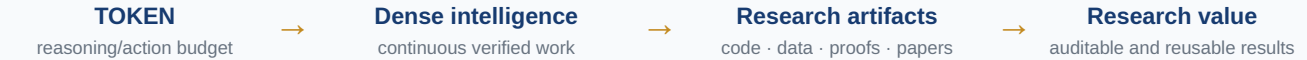
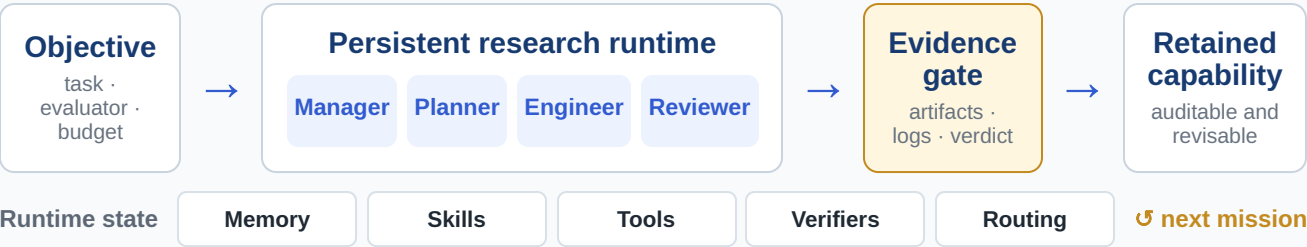
Continuous proposal—execution—verification over durable state

Invention

Long horizon

Verifiable

A Runtime architecture and fixed-model evolution



Mission trajectory τ_t Plan → Retrieve → Build → Experiment → Review → Distill bounded outcome · persisted evidence

$$\rho I(T) = \frac{1}{T} \int \dot{n}_{tok}(t) \eta_r(t) \eta_a(t) \eta_v(t) dt$$

density of useful reasoning, action, and verification—not token volume alone

$$H_{t+1} = U(H_t, \tau_t, E_t), \theta_{t+1} = \theta_t$$

evidence updates the runtime while model parameters remain fixed online

B Public evidence in task-native units

SOL-ExecBench website snapshot
Global #6 · 2× #1 · 7 top-3
B200 · 101 kernels
Two head-to-head wins over Recursive

nanochat · B200 artifact digest
0.9636 BPB
5 min · 1×B200 · 426 attempts
Human SOTA: 0.9646

nanochat · H100 website snapshot
0.9855 BPB
5 min · 1×H100 · 37 mechanisms
Human SOTA: 0.9879

nanoGPT speedrun artifact digest
79.77 seconds
8×H100 · N=10
Same-device human #83: 80.18s

AARRI-Bench website snapshot
63/82 · 76.8%
82 research-intern tasks
Paper-reported best: 68.3%

Arbor comparison website snapshot
28.0 gap
Math-Reasoning Data
Arbor 20.83 · Claude Code 8.33 · Codex 6.25

41 de-duplicated research artifacts across 6 programs 35 manuscripts · 6 drafts

Architecture is schematic; result cards are source-quoted under their printed protocols and are not cross-normalized.